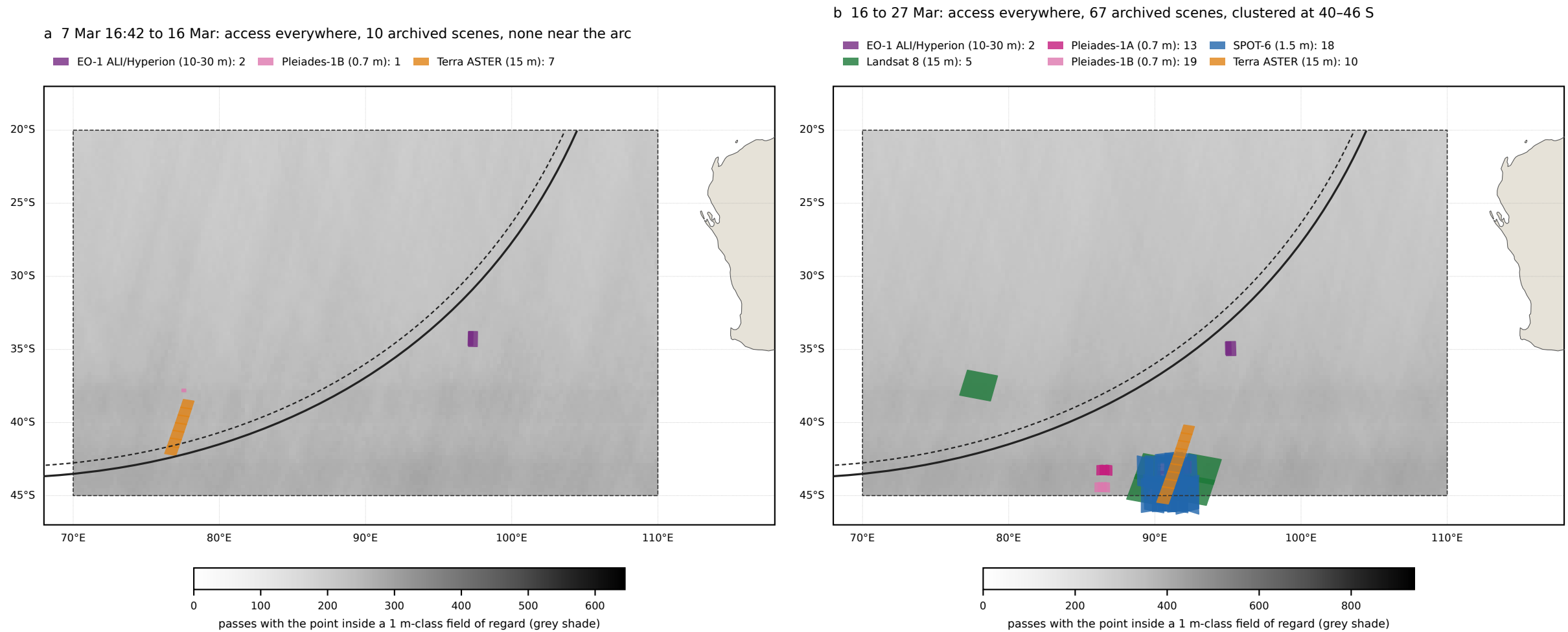
What satellites could have imaged against what the open archives hold, southern Indian Ocean, March 2014

Grey: how many usable passes (optical in daylight, radar at any time) of sensors able to resolve a 1 m object had each point inside their field of regard. Colour: footprints of scenes that exist in the open catalogues (USGS, NASA, Airbus OneAtlas); counts per satellite above each panel. Black solid: 7th arc; dashed: 6th arc; dashed box: study area 20–45 S, 70–110 E.



Plain-language note: in both periods every point of the arc area could have been imaged many times over. In the first period the open archives hold only three acquisitions in the whole area, none of them within 50 km of the arc: an ASTER strip on 8 Mar at 76–78 E (4.5 h after the last transmission, 68–100 % cloud, most likely a scheduled island/ocean pass), an EO-1 pass on 9 Mar at 34.3 S 97.4 E (230–245 km east of the arc; purpose unknown) and one routine Pléiades scene of Ile Amsterdam. In the second period 64 of the 67 scenes form one cluster at 40–46 S, 86–94 E, built up from 21 Mar, where the surface search and the satellite sightings were, 400–800 km south-west of the arc; cloud cover was above 85 % on most of them. The catalogues of RADARSAT-2, TerraSAR-X, WorldView/GeoEye, RapidEye, SPOT-5, Gaofen-1, KOMPSAT, THEOS and Cartosat need accounts the project does not hold, so these counts are a lower bound.

Technical: access raster from the SGP4 pass set (see satellite-access-counts map); scene footprints from USGS LandsatLook STAC (landsat-c2l1, eo-1-ali/hyperion-l1gst), NASA CMR AST_L1A, Airbus OneAtlas anonymous opensearch, queried 20 Oct 2026; bbox 70–110 E, 45–20 S, 7 Mar 16:00 to 28 Mar 00:00 UTC. EO-1 dates from scene-id day-of-year (times inferred); Airbus times from product identifiers. Copernicus confirmed no Sentinel-1 before Apr 2014. Scene table: archive-scenes-2014-03.csv; per-catalogue access status: archive-catalogue-report.md.